

NT 11.

$$H_0: \xi \sim p_0(x) = \begin{cases} 1, & x \in (0, 1) \\ 0, & x \notin (0, 1) \end{cases} = I_{(0, 1)}(x)$$

$$H_1: \xi \sim p_1(x) = \begin{cases} \frac{e^{1-x}}{e-1}, & x \in (0, 1) \\ 0, & x \notin (0, 1) \end{cases}$$

a) $n=1$, упростить значимость α

$$l(x) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq c$$

$$G: x \leq A$$

$$P(x \leq A | H_0) = \alpha$$

$$\int_0^A 1 dx = A = \alpha \Rightarrow \underline{G: x \leq \alpha}$$

$$\begin{aligned} W = P(x \leq A | H_1) &= \int_0^A \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1} \int_0^A e^{-x} dx = \\ &= \frac{e}{e-1} \cdot \left. e^{-x} \right|_0^A = \frac{e}{e-1} (1 - e^{-\alpha}) \end{aligned}$$

$$\alpha_1 = P(x \leq A | H_0) = \alpha$$

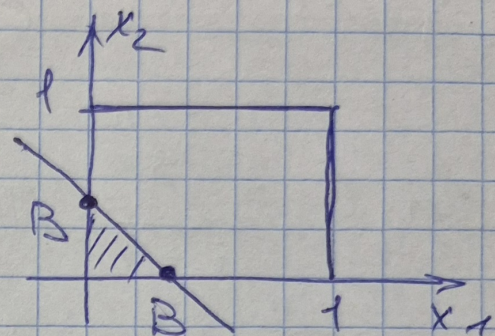
$$\alpha_2 = P(x > A | H_1) = 1 - W = \frac{e^{1-\alpha} - 1}{e-1}$$

§) Теперь $n=2$

$$f = \frac{L_1}{L_2} = \frac{e^{1-x_1} e^{1-x_2}}{(e-1)^2} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq e$$

$$G: x_1 + x_2 \leq B$$

$$P(x_1 + x_2 \leq B | H_0) = \alpha$$



$$\iint 1 \, dx_1 \, dx_2 = \alpha$$

$$\left. \begin{array}{l} 0 \leq x_1 \leq 1 \\ 0 \leq x_2 \leq 1 \\ x_1 + x_2 \leq B \end{array} \right\} E$$

$$\frac{B^2}{2} = \alpha \Rightarrow B = \sqrt{2\alpha} \leq 1$$

$$\alpha \leq \frac{1}{2}$$

$$\Rightarrow G: x_1 + x_2 \leq \sqrt{2\alpha}$$

$$W = P(x_1 + x_2 \leq B | H_1) = \int \int \frac{e^{1-x_1} e^{1-x_2}}{(e-1)^2} \, dx_1 \, dx_2 =$$

$$= \int_0^B \frac{e^2}{(e-1)^2} e^{-x_1} \, dx_1 \int_0^{B-x_1} e^{-x_2} \, dx_2 = \int_0^B \frac{e^2}{(e-1)^2} e^{-x_1} \cdot (1 - e^{x_1-B}) \, dx_1$$

$$= \int_0^B \frac{e^2}{(e-1)^2} (e^{-x_1} - e^{-B}) \, dx_1 = \left(\frac{e}{e-1} \right)^2 [1 - e^{-B} - e^{-B} \cdot B]$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = 1 - W$$

с) ассимпт. кр. по выборке n .

$$l(\vec{x}_n) = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} = \frac{\prod_{i=1}^n P_1(x_i)}{\prod_{i=1}^n P_0(x_i)} \geq c$$

$$\ln l = \sum_{i=1}^n \ln \frac{P_1(x_i)}{P_0(x_i)} \geq \ln c$$

$$\sum_{i=1}^n (1 - x_i) - n \ln(e-1) \geq \ln c$$

$$\Rightarrow G: \sum_{i=1}^n x_i \leq A$$

$$\bar{x} \leq B$$

$$P(\bar{x} \leq B | H_0) = \alpha$$

$$\text{УПТ: } \frac{\bar{x} - \mu_{\xi}}{\sqrt{D_{\xi}}} \sqrt{n} \sim N(0, 1)$$

$$H_0: \mu_{\xi} = \frac{1}{2}$$

$$D_{\xi} = \frac{1}{12}$$

$$P(\bar{x} \leq B | H_0) = P\left(\frac{\bar{x} - \frac{1}{2}}{\frac{\sqrt{1/12}}{\sqrt{n}}} \sqrt{n} \leq \frac{B - \frac{1}{2}}{\frac{\sqrt{1/12}}{\sqrt{n}}} \sqrt{n}\right)$$

$$G: \bar{x} \leq \frac{1}{2} + \frac{u_{\alpha}}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B | H_1)$$

$$H_1: M_g = \int_0^1 x p_1(x) dx = \int_0^1 \frac{x e^{1-x}}{e-1} dx = \frac{e-2}{e-1} = 0,418$$

$$M_g^2 = \frac{e}{e-1} \int_0^1 x^2 e^{-x} dx = \frac{2e-5}{e-1}$$

$$D_g = M_g^2 - (M_g)^2 = \frac{(2e-5)(e-1) - (e-2)^2}{(e-1)^2} = \frac{e^2 - 3e + 1}{(e-1)^2}$$

УПТ:

$$W = P\left(\frac{\bar{x} - M_g}{\sqrt{D_g}} \sqrt{n} \leq \frac{B - M_g}{\sqrt{D_g}} \sqrt{n}\right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$\gamma = \frac{\frac{1}{2} + \frac{U\alpha}{\sqrt{12n}} - 0,418}{\sqrt{D_g}} \sqrt{n} = \frac{U\alpha}{\sqrt{12}} + 0,082\sqrt{n}$$

, т.е. $W \xrightarrow[n \rightarrow \infty]{} 1$ $\xrightarrow[n \rightarrow \infty]{} +\infty$
 K_p -й составитель

$$\begin{cases} \alpha_1 = \alpha \\ \alpha_2 = 1 - \alpha \end{cases}$$

$$d) \quad x_{\min} < C$$

$$\begin{cases} \alpha_1 = \alpha \\ \alpha_2 \rightarrow \min \end{cases}$$

$$\alpha_1 = P(x_{\min} < C / H_0) = 1 - (1 - F(c))^n \leq \alpha$$

$$(\text{Значит } F(x) = \int_{-\infty}^x p_0(t) dt)$$

$$(1 - F(c))^n \geq 1 - \alpha$$

$$1 - F(c) \geq \sqrt[n]{1 - \alpha} \Rightarrow C \leq 1 - \sqrt[n]{1 - \alpha}$$

$$\begin{aligned} \alpha_2 &= P(x_{\min} \geq C / H_1) = 1 - P(x_{\min} < C / H_1) \\ &= 1 - 1 + (1 - \Phi(c))^n = (1 - \Phi(c))^n \rightarrow \min \end{aligned}$$

$$\text{где } \Phi(c) = \int_{-\infty}^c p_1(x) dx.$$

$$\begin{aligned} \frac{\partial \alpha_2}{\partial c} &= n(1 - \Phi(c))^{n-1} \cdot (-p_1(c)) = \\ &= -\frac{ne^{1-c}}{e-1} (1 - \Phi(c)) \end{aligned}$$

$$\Phi(c) = \int_0^c p_1(x) dx = \frac{e}{e-1} \int_0^c e^{-x} dx =$$

$$z \frac{e(1-e^{-c})}{e-1}$$

$$\Rightarrow 1 - \varphi(c) z \frac{e^{1-c} - 1}{e-1}$$

$$\Rightarrow \frac{\partial d_2}{\partial c} z \frac{-ne^{1-c}}{e-1} \left(\frac{e^{1-c} - 1}{e-1} \right)^{n-1} < 0$$

где $c \in (0, 1)$

мин на $C \leq 1 - \sqrt[n]{1-\alpha}$ достигается
при $C = 1 - \sqrt[n]{1-\alpha}$

$$G: x_{\min} = 1 - \sqrt[n]{1-\alpha}$$

$$\Rightarrow d_2 = (1 - \varphi(1 - \sqrt[n]{1-\alpha}))^n =$$

$$z \left[\frac{1}{e-1} (e^{1-1+\sqrt[n]{1-\alpha}} - 1) \right]^n = \left(\frac{e^{\frac{n\sqrt[n]{1-\alpha}}{n}} - 1}{e-1} \right)^n =$$

$$z \left(\frac{e \left(1 + \frac{1}{n} \ln(1-\alpha) \right) + o\left(\frac{1}{n}\right) - 1}{e-1} \right)^n =$$

$$z \left(1 + \frac{e \ln(1-\alpha)}{(e-1)n} + o\left(\frac{1}{n}\right) \right)^n$$

$$\Rightarrow \lim_{n \rightarrow \infty} d_2 = \lim_{n \rightarrow \infty} \left(1 + \frac{e \ln(1-\alpha)}{e-1} \cdot \frac{1}{n} \right)^n =$$

$$= e^{\frac{e \ln(1-\alpha)}{e-1}} = (1-\alpha)^{\frac{e}{e-1}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} W = \lim_{n \rightarrow \infty} (1 - \alpha_2) = 1 - (1 - \alpha) \frac{e}{e-1} \neq 1$$

$\Rightarrow$ кр. не экв. состоятельны

$$\begin{aligned} \alpha_1 &= 1 - (1 - F(1 - \sqrt[n]{1 - \alpha}))^n = \\ &= 1 - (\sqrt[n]{1 - \alpha})^n = \alpha \end{aligned}$$

$$\alpha_1 = \alpha$$

$$\alpha_2 = \left(\frac{e^{\sqrt[n]{1 - \alpha}} - 1}{e - 1} \right)^n$$